<u>SCIENCE</u> -BPT

<u>14-01-24</u>

Q1. 3) Deviate towards Normal

2. 4) IV

3. 3) Scattering of lights

4. 3) Ciliary muscles

5. 4) Assertion is false and Reason is true

6. The refractive index of a transparent medium is the change of speed ~~of~~ light when it enter this medium to or from another medium.

Here,

refractive index of medium = 1.5

speed of light ⇒ 2×10^8 (in that medium)

we know,

$$n = \frac{c}{v}$$ (where, v - speed in medium
c - speed in vaccum)

so, $$1.5 = \frac{c}{2 \times 10^8} \rightarrow c = \frac{\cancel{15} \times 2 \times 10^8}{\cancel{10}}$$

$$= 3 \times 10^8 \ ms^{-1}$$

7. The fuse is a safety device which is used in series with electrical appliance as it has a specific value of current flow so if the current value exceeds that value then it will melt and break the circuit and saves devices from damage.

A fuse with defined rating allows only that safest amount of current to flow that cannot do any damage to device and if we replace with larger rating then the increased amount of current will cause damage to devices

8. Here,

for that person

$u = -25$

$v = -75$

using lens formula = $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\Rightarrow \frac{1}{f} = \frac{-1}{75} + \frac{1}{25} \Rightarrow \frac{-1+3}{75} \Rightarrow \frac{2}{75}$

$f = \frac{75}{2} = 37.5\ cm$

As the focal length has positive value indicates it is a convex lense

$\begin{array}{r} 25 \\ \times 6 \\ \hline 150 \end{array}$ Date $\begin{array}{r} 25 \\ \times 3 \\ \hline 225 \end{array}$
Page

$pauer = \frac{1}{f} = \frac{200}{75} \rightarrow \cdot 026$

9. we have wire length $= l$
resistance $= R \rightarrow R = \rho \frac{l}{A}$ — i
then,
new length $= 2l$
new area $= \frac{A}{2}$

New Resistance $= \rho \frac{l}{A} \rightarrow \rho \frac{2l}{\frac{A}{2}} \Rightarrow \rho \frac{4l}{A}$

$\Rightarrow 4 \times \rho \frac{l}{A}$

from eq i $\Rightarrow 4 \cdot R$

i) The resistance will increase by four times
ii) The resistivity will not increase as resistivity is the materilistic property it changes only with diffrent material, physical factor doesn't change the resistivity as streching, doubling.

10. Using the right thumb rule,
we will get that
at P $\rightarrow$ The direction of magnetic fields are in the plane of paper

at Q - The direction of magnetic feild is autward the plane of paper
we know that the strngth of magneticc feild is inversely proptional to the distance from wire
hence,

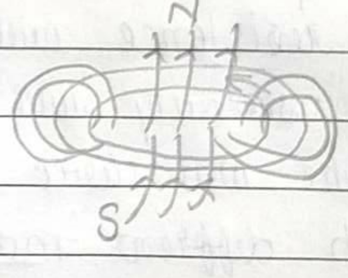
as $r_1 > r_2$

$P > Q$

→ The strngth is greater Q than P

ii) a) The direction of magntic feild determined through right thumb rule

~~ii)~~ b) At the centre of current carrying circular loop, the direction of magnetic feild is from dawnward to upward as straight line



11. i) Here,

we know in parallel devices voltage flows equally across each
and,

$$R = \frac{V}{I}$$

then,

$10\Omega = \frac{3}{I}$, $I_1 = \frac{3}{10}$

$15\Omega = \frac{3}{I_2}$, $I_2 = \frac{3}{15}$

~~Total current~~ $= \cancel{\frac{3}{10}}$ ~~+~~ .

ii) The nature is a straight line graph

iii) For series combination,

The ~~sum~~ resultant is grtearn than the indivuals and her,

$C > B$, $C > A$, $B > A$

It shows C is series combination of other two

12. i) $h_1 = 4$

$u = -25$

$f = -15$ (as it concave mirror)

i) v - using mirror formula,

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \rightarrow \frac{-1}{15} = \frac{1}{v} - \frac{1}{25}$

$\frac{1}{v} = \frac{1}{25} - \frac{1}{15} \Rightarrow \frac{3-5}{75} = \frac{-2}{75}$

$v = -\frac{75}{2} \Rightarrow -37.5$ cm

ii) we know

$$\frac{-v}{u} = \frac{h_2}{h_1}$$

$$\Rightarrow \frac{\cancel{8}-(-75)}{2\times -25} \Rightarrow \frac{h_2}{4}$$

$$\rightarrow \frac{\cancel{75}\times\cancel{4}}{25\times 2} \Rightarrow 3\times 2 = 6\text{ cm} = h_2$$

The size of image is 2 cm more than that of object (or in large)

iii)